

EDUCATION

- **Iowa State University** Ames, IA
B.S. in Mathematics *August 2018 – May 2021*
- **University of Utah** Salt Lake City, UT
Ph.D. in Mathematics (expected) *August 2021 – Present*
 - **Advisor:** Tommaso de Fernex

PUBLICATIONS

- [Mer26] Zach Mere, *A change-of-variables formula for motivic integration on Greenberg schemes* (2026). In preparation.
- [dFM] Tommaso de Fernex and Zach Mere, *Relative Mather discrepancy on arc spaces*, Michigan Mathematical Journal. Accepted for publication; arXiv:2508.12420 (2025).
- [MM22] Jason McCullough and Zach Mere, *G-quadratic, LG-quadratic, and Koszul quotients of exterior algebras*, Communications in Algebra **50** (2022), no. 8, 3284–3300.

AWARDS

- **NSF-RTG Fellowship (DMS-1840190):** Fall 2023
- **Gertrude Herr Adamson Award:** Spring 2020
- **National Merit Scholarship (Semifinalist):** Spring 2017

TALKS

- **University of Utah** Salt Lake City, UT
Algebraic Geometry Seminar *Fall 2025 – Present*
 - **Change-of-variables in motivic integration:** Dec. 2025
- **University of Utah** Salt Lake City, UT
Basic Algebraic Geometry Eating and Learning Seminar (BAGELS) *Fall 2023 – Present*
 - **A Witt vector version of Greenberg’s theorem:** Mar. 2026
 - **Greenberg’s approximation theorem:** Feb. 2026
 - **The change-of-variables formula in motivic integration:** Sep. 2025
 - **Differentials on arc spaces:** Feb. 2025
 - **A friendly introduction to arc spaces:** Oct. 2024
 - **Arc spaces and the Nash problem:** Feb. 2024
 - **Fujita’s vanishing theorem:** Nov. 2023

TEACHING

- **University of Utah** Salt Lake City, UT
Instructor of record *May 2022 – Present*
 - **Summer 2026:** MATH 1320: Engineering Calculus II
 - **Fall 2025:** MATH 1050: College Algebra
 - **Summer 2025:** MATH 2200: Discrete Mathematics
 - **Spring 2025:** MATH 1320: Engineering Calculus II
 - **Fall 2024:** MATH 2250: Differential Equations and Linear Algebra
 - **Summer 2024:** MATH 1320: Engineering Calculus II
 - **Summer 2023:** MATH 1320: Engineering Calculus II
 - **Spring 2023:** MATH 1030: Introduction to Quantitative Reasoning
 - **Fall 2022:** MATH 1090: Business Algebra
 - **Summer 2022:** MATH 1320: Engineering Calculus II

- **University of Utah**

Teaching assistant

- **Spring 2026:** MATH 1320: Engineering Calculus II
- **Spring 2022:** MATH 1320: Engineering Calculus II
- **Fall 2021:** MATH 1320: Engineering Calculus II

Salt Lake City, UT

Aug 2021 – May 2022